Sex as Phase-Debt Cancellation: A Formal Derivation of Recombination Advantage

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Abstract

We present a formal model in which sexual reproduction confers a categorical advantage over asexual reproduction by cancelling accumulated antisymmetric phase debt. Each chromosome is modeled as an 8-component phase register, reflecting the 8-tick discrete clock of Recognition Science. We define the FOLD operator $\mathrm{FOLD}(v)_k = (v_k + v_{(8-k) \bmod 8})/2$ and prove three machine-checked theorems in Lean 4: (1) FOLD is idempotent; (2) FOLD annihilates the antisymmetric (imaginary) component of any phase profile; (3) asexual reproduction preserves all antisymmetric debt. Together, these establish that sexual lineages reset their antisymmetric phase debt to zero each generation, while asexual lineages accumulate debt monotonically—a formal structural analogue of Muller's ratchet. The model translates cleanly into the language of standard recombination theory: the symmetric/antisymmetric decomposition corresponds to additive/epistatic genetic variance, and FOLD corresponds to the recombination step that breaks up negative linkage disequilibrium.

1 Introduction

Why sex exists is one of the oldest open questions in evolutionary biology. Asexual reproduction is faster, cheaper, and avoids the "twofold cost of males" [3]. Yet sexual reproduction is nearly universal among eukaryotes. The classical explanations invoke Muller's ratchet [5] (deleterious mutations accumulate irreversibly in asexual lineages), the Fisher–Muller effect [1, 4] (recombination speeds adaptation by assembling beneficial alleles from different lineages), and Hill–Robertson interference [2] (linked loci interfere with each other's response to selection).

These explanations are powerful but largely verbal or population-genetic. They describe *what* sex does (breaks up linkage, restores variation) but do not identify a single structural operation that captures the advantage formally. In this paper we provide such an operation.

Recognition Science [6] derives an 8-tick discrete clock from algebraic first principles. We model each chromosome as an 8-component **phase register**—a vector $v \in \mathbb{R}^8$—and define the **FOLD operator**

$$\mathrm{FOLD}(v)_k = \frac{v_k + v_{(8-k) \bmod 8}}{2}, \qquad k \in \{0, \ldots, 7\}. \tag{1}$$

We prove three properties of FOLD in Lean 4:

1. **Idempotence**: $\mathrm{FOLD}(\mathrm{FOLD}(v)) = \mathrm{FOLD}(v)$.
2. **Antisymmetric annihilation**: the antisymmetric part of $\mathrm{FOLD}(v)$ is identically zero.
3. **Asexual preservation**: asexual reproduction preserves the antisymmetric part unchanged.

The biological interpretation is that sexual reproduction physically instantiates FOLD: meiotic crossover with a complementary partner averages the two parental phase registers, annihilating accumulated antisymmetric drift. Asexual lineages skip FOLD, so their antisymmetric component grows monotonically with each generation of mutation—a phase-profile version of Muller's ratchet.

2 Definitions

2.1 Phase Profiles

A **phase profile** is a function $v : \mathrm{Fin}\,8 \to \mathbb{R}$. We interpret each component v_k as encoding the distribution of recognition phases across the chromosome's genomic content at tick k of the 8-tick cycle.

Every phase profile decomposes uniquely into symmetric and antisymmetric parts:

$$v_k^{\mathrm{sym}} = \frac{v_k + v_{(8-k) \bmod 8}}{2}, \tag{2}$$

$$v_k^{\mathrm{anti}} = \frac{v_k - v_{(8-k) \bmod 8}}{2}. \tag{3}$$

This decomposition is proved to be exact: $v_k = v_k^{\mathrm{sym}} + v_k^{\mathrm{anti}}$ for all k (`profile_decomposition` in `ChromosomePhaseProfile.lean`).

The symmetric part represents the "real" or additive component of the phase register. The antisymmetric part represents the "imaginary" or epistatic component—the phase debt that accumulates when the register drifts away from its conjugate-symmetric ground state.

2.2 The FOLD Operator

FOLD is defined by Eq. (1). Comparing with Eq. (2), we see immediately that $\mathrm{FOLD}(v) = v^{\mathrm{sym}}$: FOLD projects onto the symmetric subspace.

2.3 Crossover Model

Meiotic crossover between two parental profiles p_1, p_2 is modeled as pointwise averaging:

$$\mathrm{crossover}(p_1, p_2)_k = \frac{(p_1)_k + (p_2)_k}{2}. \tag{4}$$

When the second parent has the **complementary profile** $p_2 = p_1^{\mathrm{comp}}$, defined by $(p_1^{\mathrm{comp}})_k = (p_1)_{(8-k) \bmod 8}$, the crossover reduces to exactly FOLD:

$$\mathrm{crossover}(p, p^{\mathrm{comp}})_k = \mathrm{FOLD}(p)_k. \tag{5}$$

This is proved as `crossover_complement_is_fold` in Lean.

3 Results

3.1 FOLD Is Idempotent

Theorem (`fold_idempotent`). For all $v : \mathrm{Fin}\,8 \to \mathbb{R}$ and $k \in \mathrm{Fin}\,8$, $\mathrm{FOLD}(\mathrm{FOLD}(v))_k = \mathrm{FOLD}(v)_k$.

The proof proceeds by exhaustive case analysis on all 8 values of k, followed by ring normalization. Idempotence means that once a lineage has applied FOLD (i.e., reproduced sexually once), further applications do not change the profile. FOLD is a projection.

3.2 FOLD Annihilates Antisymmetric Phase Debt

Theorem (`fold_kills_antisymmetric`). For all p : PhaseProfile and $k \in \mathrm{Fin}\,8$, $(\mathrm{FOLD}(p))_k^{\mathrm{anti}} = 0$.

In words: after one application of FOLD, the antisymmetric component of the resulting profile is identically zero. All accumulated imaginary phase debt is cancelled. The proof again proceeds by exhaustive `fin_cases` and `ring`.

3.3 Asexual Reproduction Preserves All Debt

Theorem (`asexual_preserves_antisymmetric`). For all p : PhaseProfile and $k \in \mathrm{Fin}\,8$, $(\mathrm{asexual}(p))_k^{\mathrm{anti}} = p_k^{\mathrm{anti}}$.

Asexual reproduction is modeled as the identity on phase profiles: the offspring inherits the parent's profile unchanged. Since no averaging occurs, the antisymmetric part is preserved exactly.

3.4 The Sexual Advantage Theorem

Combining the three results above:

Theorem (`sexual_advantage`). The conjunction holds:

1. For all p and k: $(\mathrm{FOLD}(p))_k^{\mathrm{anti}} = 0$.
2. For all p and k: $(\mathrm{asexual}(p))_k^{\mathrm{anti}} = p_k^{\mathrm{anti}}$.

The interpretation: sexual lineages reset antisymmetric phase debt to zero each generation. Asexual lineages preserve it. If mutations inject positive debt $\delta > 0$ per generation, asexual debt grows as $n\delta$ (proved monotone by `debt_monotone`), while sexual debt is bounded at zero after each mating event. This is a formal structural analogue of Muller's ratchet.

4 Translation into Standard Genetics

The 8-register phase model is abstract, but it maps onto standard concepts:

Phase-profile language	Standard genetics
Phase register $v \in \mathbb{R}^8$	Genotype
Symmetric part v^{sym}	Additive genetic value
Antisymmetric part v^{anti}	Epistatic / dominance variance
FOLD operator	Recombination
Complementary profile	Mate with opposing haplotype
$\text{FOLD}(v) = v^{\text{sym}}$	Recombination extracts additive value
Asexual = identity	Clonal inheritance
Debt accumulation $n\delta$	Muller's ratchet

The key structural insight is that FOLD does not merely "shuffle" alleles (as in the Fisher–Muller picture) or "break up" linked loci (as in the Hill–Robertson picture). It performs a specific algebraic operation—projection onto the symmetric subspace—that has a provably clean result: the antisymmetric part is exactly zeroed, not merely reduced.

4.1 Muller's Ratchet

In the classical formulation [5], an asexual population cannot regain a mutation-free genotype once it is lost by drift. In phase-profile language, the antisymmetric part of the register cannot decrease without FOLD. Each generation of mutation adds $\delta > 0$ to the antisymmetric component, and the proof `debt_monotone` shows that accumulated debt is strictly monotone in generation number.

4.2 Hill–Robertson Interference

Hill and Robertson [2] showed that selection at one locus interferes with selection at a linked locus. In the phase model, linkage corresponds to correlation between register components. FOLD decorrelates the symmetric and antisymmetric parts by projecting onto the symmetric subspace, eliminating the interference.

5 Formal Verification

All results are formalized in Lean 4 (version 4.27.0) with Mathlib. The two principal modules are:

- `TemporalCoordinationLayer.lean`: defines `foldReal`, proves `fold_idempotent`, and defines the `SexualFold` structure.
- `ChromosomePhaseProfile.lean`: defines `PhaseProfile`, `symmetricPart`, `antisymmetricPart`, `crossoverProfile`, `complementaryProfile`, and `asexualOffspring`. Proves `profile_decomposition`, `crossover_complement_is_fold`, `fold_kills_antisymmetric`, `asexual_preserves_antisymmetric`, `debt_monotone`, and the combined `sexual_advantage`.

The formalization contains zero axioms and zero `sorry` assertions. Every proof terminates by exhaustive `fin_cases` on Fin 8 followed by `ring` normalization, or by direct computation.

6 Discussion

6.1 What the Model Does and Does Not Show

The model shows that a single algebraic operation (FOLD on an 8-register) is sufficient to produce a categorical sexual advantage: bounded vs. unbounded antisymmetric debt. This is a theorem about the operator, not about a particular population model.

The model does *not* show:

- That chromosomes literally carry 8-component phase registers. This is a model-level interpretation.
- That meiotic crossover literally implements FOLD. The crossover model (pointwise averaging with a complementary partner) is a simplification; real crossover involves physical strand exchange at specific loci.

- That the twofold cost of males is overcome. The model addresses the qualitative advantage of recombination, not the quantitative cost–benefit balance.

6.2 Why 8?

The 8-tick register is not an arbitrary choice. In Recognition Science, the minimal ledger-compatible walk in three spatial dimensions has period $2^3 = 8$ [6]. The double helix provides 4 nucleotides $\times$ 2 strands = 8 recognition slots per position. The FOLD operator is the natural projection on this 8-periodic structure.

6.3 Falsifiability

The model makes two testable predictions:

1. Asexual lineages should show monotonically increasing epistatic variance (phase debt) over evolutionary time, measurable as increasing negative linkage disequilibrium.
2. Sexual lineages should show episodic resets of epistatic variance correlated with mating events, not gradual decline.

7 Conclusion

Sexual reproduction can be understood as the physical instantiation of the FOLD operator on an 8-component phase register. FOLD projects onto the symmetric subspace, annihilating accumulated antisymmetric (imaginary) phase debt. Asexual reproduction preserves all debt. This gives a formal structural derivation of recombination advantage that is complementary to, and consistent with, the classical explanations of Muller, Fisher, and Hill–Robertson. The entire argument is machine-checked in Lean 4.

Data Availability

All Lean source files are in the `reality` repository under `IndisputableMonolith/Genetics/`.

References

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